

Q3. IIR filters.

(a).

it has 3 pairs of poles :

$$s_1 = \Omega_c \cdot e^{j\frac{7}{12}\pi} \quad \frac{\pi}{2} + \frac{2k+1}{N} \cdot \frac{\pi}{2}$$

$$s_1^* = \Omega_c \cdot e^{-j\frac{7}{12}\pi} \quad \frac{2\pi}{12} = \frac{\pi}{6}$$

$$s_2 = \Omega_c \cdot e^{j\frac{9}{12}\pi} \quad \frac{\pi}{2} + \frac{1}{2} \cdot \frac{\pi}{6}$$

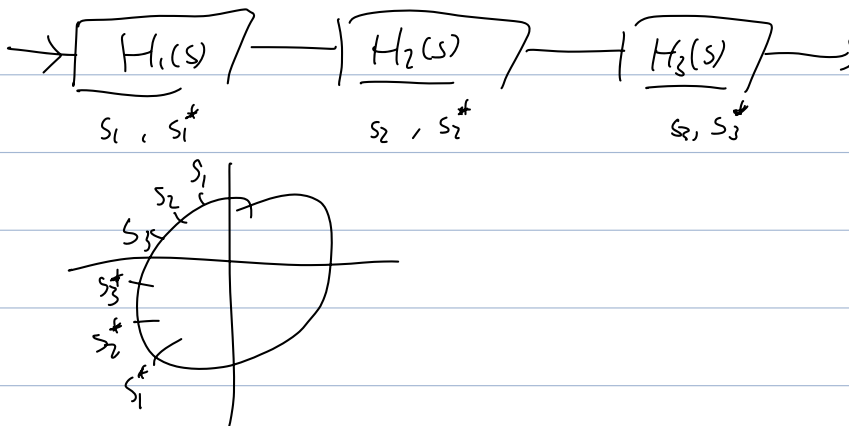
$$s_2^* = \Omega_c \cdot e^{-j\frac{9}{12}\pi} \quad \vdots$$

$$s_3 = \Omega_c \cdot e^{j\frac{11}{12}\pi}$$

$$s_3^* = \Omega_c \cdot e^{-j\frac{11}{12}\pi}$$

it can be divided into 3 2-order system :

$$H(s) = \prod_{k=1}^3 \frac{1}{(s-s_k)(s-s_k^*)}$$



(b) bilinear transform.

$$s_k = \frac{2}{T} \cdot \frac{1 - z_k^{-1}}{1 + z_k^{-1}} \Rightarrow z_k = \frac{1 + \frac{T}{2} s_k}{1 - \frac{T}{2} s_k} \Rightarrow H(s) \Rightarrow h(n)$$

The diagram shows the bilinear transform mapping from the s-plane to the z-plane. The s-plane has a horizontal axis σ and a vertical axis $j\omega$. The mapping is shown as a curve starting from the origin and approaching the $j\omega$ axis. The $j\omega$ axis is labeled with π and $-\pi$. The σ axis is labeled with ∞ and $-\infty$. The mapping is labeled $\Omega = \infty \rightarrow \omega = \pi$ and $\downarrow$ Nyquist.

* preserve stability (left plane $\rightarrow$

* it maps $\Omega = 0$ to $\omega = \pi$, so the Nyquist freq, experience the max. attenuation.

c). Yes there will be.

$$H(z) = \frac{K}{T \left(\frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}} - s_k \right)} = \frac{K (1+z^{-1})^{1c}}{T \left(\frac{2}{T} (1-z^{-1}) - s_k (1+z^{-1}) \right)}$$

$z = -1, 1$, zeros.